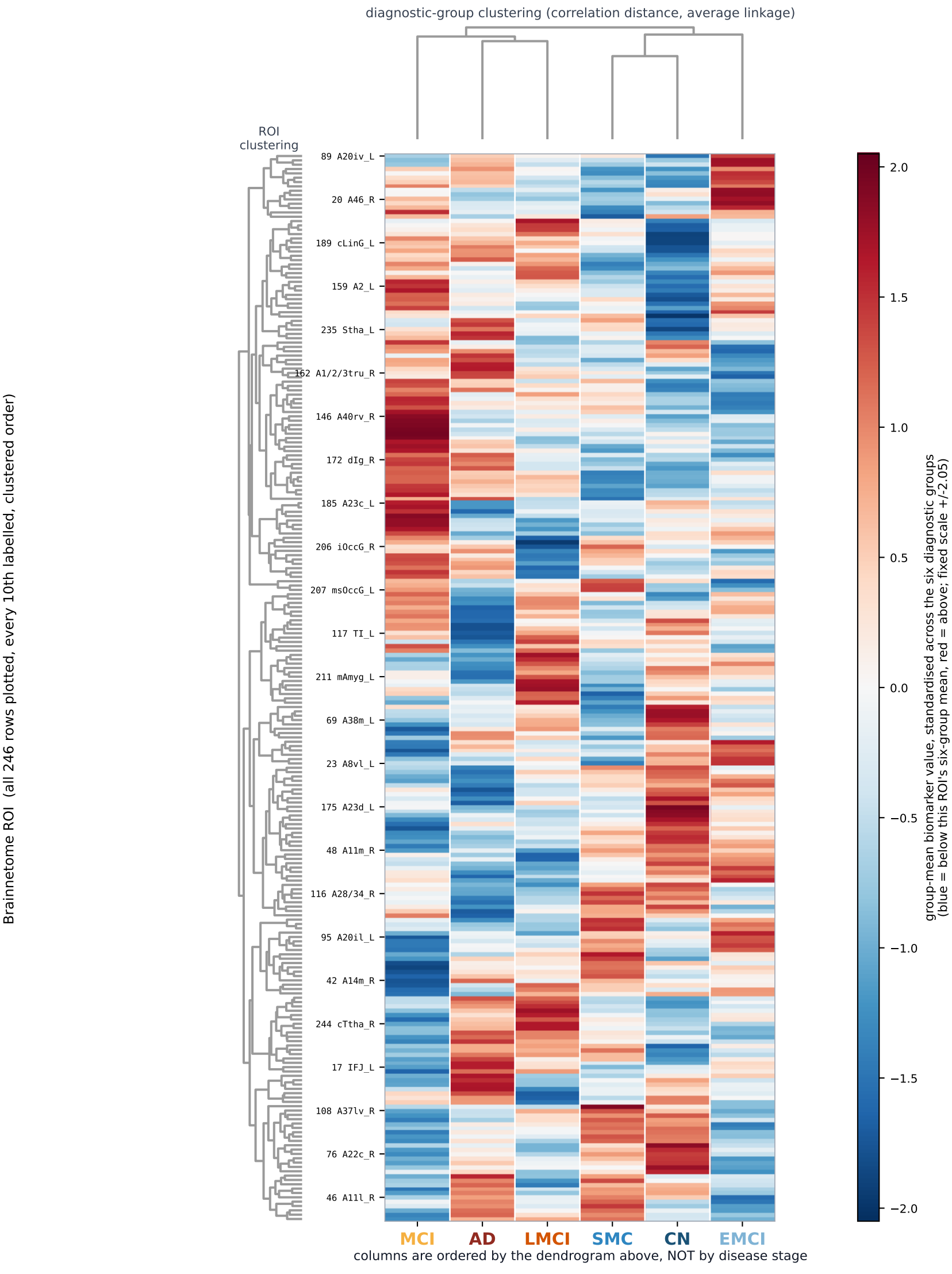
Hierarchical Clustering of Alzheimer's Diagnostic Groups
Standardised signed degree centrality (DC_z)

cells are the MEAN DC_z of the subjects in each diagnostic group (n = 20/20/20/16/20/20), standardised across the six groups per ROI (ddof = 1)
groups clustered by correlation distance + average linkage over the 246-D profile; ROIs by euclidean distance + average linkage

group P-score = -log10(q) from the MULTIVARIATE stage-vs-rest permutation test (24-test family)

max ROI P = largest ROI-level P-score for that group -- DESCRIPTIVE, not a test | both are far below the 1.301 significance threshold

	MCI	AD	LMCI	SMC	CN	EMCI
group P-score	**0.738**	**0.128**	**0.119**	**0.119**	**0.535**	**0.119**
max ROI P	0.137	0.137	0.137	0.055	0.137	0.137



clinical progression order: CN → SMC → EMCI → MCI → LMCI → AD

bootstrap stability of the branches (1000 resamples of subjects within groups): {CN,SMC} 49%; {LMCI,AD} 26%; {CN,SMC,EMCI} 19%

Heatmap shows BIOMARKER ACTIVITY; the top row shows STATISTICAL SIGNIFICANCE. They are different quantities. The tree clusters on biomarker activity.

Descriptive only. Branch heights are distances, not p-values. No group reaches q < 0.05 and no ROI x stage test in this project does either.